HP at BAREC Shared Task 2026: Parameter-Efficient Ordinal Regression and Threshold Optimization for Arabic Readability

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Abstract

This paper describes the system submitted by team HP for the BAREC Shared Task 2026 on Arabic readability assessment. Driven by the Quadratic Weighted Kappa (QWK) evaluation metric, which heavily penalizes distant misclassifications, we reframe the 19-level task as an ordinal regression problem rather than standard multi-class classification. Our approach leverages pre-trained language models (AraBERTv02, ARBERT) optimized via Parameter-Efficient Fine-Tuning (LoRA) and trained with a COnsistent RAnk Logits (CORAL) loss framework incorporating sample-level class weights. To ensure robust generalization, we consolidate the available datasets and implement a 5-fold stratified cross-validation pipeline. Finally, to optimize the QWK metric on the official blind test, we apply an 18-threshold Nelder-Mead optimizer calibrated exclusively on blended Out-Of-Fold logits, yielding a blind test QWK of 0.837. Our solution is released in the Github [1].

1 Introduction

Assessing the readability of text is a crucial task in Natural Language Processing (NLP) that facilitates educational tools, text simplification systems, and accessible content generation. For Arabic, this is challenging due to its rich morphology, dialectal variations, and a scarcity of fine-grained datasets. The BAREC Shared Task 2026 addresses this gap by challenging participants to predict Arabic sentence readability across 19 finely graded levels (Elmadani et al., 2026a, 2025a,b; Habash et al., 2025).

Most traditional approaches model readability assessment as a standard multi-class classification problem, implicitly optimizing for exact match accuracy. However, this formulation is highly suboptimal when the primary evaluation metric is Quadratic Weighted Kappa. Because QWK quadratically penalizes predictions that are far from the true label, a model must respect the relative ordering of the 19 classes. To directly align the learning signal with this metric, our system approaches the task as an ordinal regression problem. Through rigorous ablation studies comparing nominal classification and standard regression baselines, we empirically demonstrate the superiority of this ordinal framing.

Furthermore, while pre-trained language models (PLMs) have achieved state-of-the-art results, full fine-tuning is computationally expensive and prone to overfitting on imbalanced datasets. We address this by leveraging Parameter-Efficient Fine-Tuning, specifically Low-Rank Adaptation (Hu et al., 2021). To maximize labeled data utilization, we consolidate all provided splits into a 5-fold stratified cross-validation pipeline. Finally, to counteract conservative prediction behavior on unseen data, we employ an 18-threshold Nelder-Mead optimizer calibrated exclusively on blended Out-Of-Fold logits.

2 Background

2.1 Task Setup and Dataset

The BAREC Shared Task 2026 focuses on fine-grained Arabic readability assessment at the sentence level (Elmadani et al., 2026a). The primary dataset provided is the BAREC Readability Corpus (Elmadani et al., 2025b; Altarbouch et al., 2025; Elmadani et al., 2026b), which contains Arabic sentences annotated across 19 progressive readability levels based on strict annotation guidelines (Habash et al., 2025).

Our team participated in the **Strict Track**, which mandates that models be trained exclusively on the training set of the BAREC Corpus. Given the strict prohibition of external corpora, linguistic lexicons, or additional data augmentation, we focused our efforts on maximizing the potential of the provided

[1] https://github.com/phonghoangKHDL/BAREC-HP-2026

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data. We achieved this by fundamentally reframing the learning signal through ordinal regression rather than relying on external knowledge. Prior work in Arabic readability modeling has explored various feature engineering and neural architecture strategies (Liberato et al., 2024; Hazim et al., 2022).

2.2 Ordinal Regression and the QWK Metric

The primary evaluation metric for this shared task is the Quadratic Weighted Kappa, which measures the agreement between predicted and actual readability levels. Unlike standard accuracy, QWK is designed to heavily penalize distant misclassifications. Given $N = 19$ classes, the penalty weight $w_{i,j}$ between a predicted level i and an actual level j is defined quadratically as:

$$w_{i,j} = \frac{(i-j)^2}{(N-1)^2} \quad (1)$$

Under this metric, confusing a Level 1 text with Level 5 incurs a penalty 16 times greater than confusing it with Level 2.

Standard multi-class classification objectives, such as Cross-Entropy or Focal Loss (Lin et al., 2018), assume all classes are mutually independent. They penalize a model identically regardless of whether its prediction is off by one level or eighteen levels, making them inherently mismatched with the QWK objective. On the other hand, standard regression losses (e.g., Mean Absolute Error) account for continuous distance but incorrectly assume that intervals between readability levels are perfectly equidistant, which rarely reflects true linguistic complexity.

Ordinal regression overcomes these limitations by acknowledging progressive difficulty without enforcing uniform spacing. By formulating the problem as a sequence of hierarchical binary boundaries via the CORAL framework (Cao et al., 2020), the model learns to respect rank order. This constraint naturally prevents extreme misclassifications, directly aligning the learning signal with the quadratic penalty mechanism of the QWK metric.

3 System Overview

Our proposed system architecture integrates deep text cleaning, Parameter-Efficient Fine-Tuning, and an ordinal loss framework to address the specific challenges of the 19-level readability task.

3.1 Data Preprocessing and Contradiction Resolution

Real-world datasets often contain noise that can severely degrade the performance of fine-grained ordinal models. We applied a rigorous data cleaning pipeline. First, for **Text Normalization**, we utilized the `ArabertPreprocessor` (Antoun et al., 2020) to strip HTML tags, URLs, extra whitespaces, and Tatweel characters, ensuring standardized inputs (Elmadani et al., 2025b). Second, for **Label Contradiction Resolution**, we identified that identical sentences were occasionally assigned to different readability levels (hard contradictions). We discarded these conflicting instances to prevent optimization instability. Furthermore, exact text duplicates were removed to mitigate data leakage and accelerate training convergence.

3.2 Model Architecture and PEFT

We employed Arabic pre-trained language models, primarily AraBERTv02 (Antoun et al., 2020) and ARBERTv2 (Abdul-Mageed et al., 2021), as our base encoders. Due to the relatively limited size of the training corpus and the vast parameter space of these models, full fine-tuning risks overfitting.

To counteract this, we integrated Low-Rank Adaptation (Hu et al., 2021). We injected trainable rank decomposition matrices into the `query`, `value`, and `dense` modules of the transformer layers. With hyperparameters set to rank $r = 16$, $\alpha = 32$, and a dropout of 0.05, we updated only a fraction ($\sim$1.7%) of the total parameters, significantly enhancing training stability and generalization.

3.3 COnsistent RAnk Logits Framework

To enforce ordinality, we replaced the standard classification head with the CORAL framework (Cao et al., 2020). Instead of predicting probabilities across 19 independent classes, the model outputs 18 binary logits. Each logit represents a binary classifier answering the question: *"Is the readability level strictly greater than k?"* for $k \in \{0, 1, \ldots, 17\}$.

Given a sentence x_i with a true label mapped to $y_i \in \{0, \ldots, 18\}$, we generate a binary target vector T_i of length 18, where $T_{i,k} = 1$ if $y_i > k$, and 0 otherwise. The model outputs 18 logits $\hat{y}_i$. The loss for a single sample is computed as the

mean of 18 Binary Cross-Entropy (BCE) losses:

$$\mathcal{L}_{CORAL}^{(i)} = \frac{1}{18} \sum_{k=0}^{17} BCE(\hat{y}_{i,k}, T_{i,k}) \quad (2)$$

During inference, the final predicted level is obtained by applying a threshold (e.g., 0) to the 18 binary logits, summing the positive responses to obtain a value in $\{0, \ldots, 18\}$, and then adding 1 to obtain the final predicted level in $\{1, \ldots, 19\}$.

3.4 Sample-level Class Weights

The BAREC dataset exhibits class imbalance, which can bias the binary classifiers in the CORAL framework (Cao et al., 2020). We computed balanced class weights W based on the training distribution. To prevent gradient explosion, these weights were clipped to the range $[0.5, 5.0]$ and re-normalized. The final loss applies these weights at the sample level:

$$\mathcal{L}_{Final} = \frac{1}{N} \sum_{i=1}^{N} W_{y_i} \cdot \mathcal{L}_{CORAL}^{(i)} \quad (3)$$

4 Experimental Setup

4.1 Two-Phase Evaluation Strategy

To ensure robust generalization and prevent leaderboard overfitting, our experimental workflow was divided into two distinct phases. In the **Development Phase**, we first evaluated AraBERTv02 (Antoun et al., 2020) under various loss objectives using standard data splits, identifying the CORAL framework as optimal. We additionally evaluated ARBERTv2 (Abdul-Mageed et al., 2021) under the identical CORAL objective on the development set to serve as an architectural baseline. We then implemented a 5-Fold Stratified Cross-Validation pipeline with multi-seed training to maximize data utilization, generating Out-Of-Fold predictions that significantly boosted our internal validation score. In the **Testing Phase** (for the blind test), we maintained this proven OOF methodology. All post-processing hyperparameters: the multi-seed blend weight and the 18 Nelder-Mead thresholds (Nelder and Mead, 1965) are finalized exclusively on the OOF logits prior to any blind test inference. The blind test set is loaded and predicted exactly once using the frozen model checkpoints and frozen thresholds.

4.2 Implementation Details

We utilized the Hugging Face `transformers` (Wolf et al., 2020) and `peft` libraries. Input sentences were tokenized with a maximum sequence length of 128. Models were trained for 5 epochs using a learning rate of 3×10^{-4} with a weight decay of 0.01. To accommodate memory constraints while maintaining a sufficient batch size, we used a per-device batch size of 8 with 4 gradient accumulation steps. All ablation experiments shared this identical configuration to ensure a fair comparison. Training was conducted using the `bfloat16` precision format, which provided improved numerical stability for the gradients, particularly for our primary CORAL loss framework.

5 Post-Processing

Our OOF experiments revealed that fine-tuned models trained on ordinal datasets often exhibit conservative prediction behaviors, clustering around majority classes to minimize expected loss under QWK. To address this, we employ the Nelder-Mead algorithm (Nelder and Mead, 1965) to jointly optimize the 18 binary decision thresholds of the CORAL framework, directly maximizing QWK on the blended multi-seed OOF logits.

Specifically, we first search for the optimal blend weight $w \in [0, 1]$ across seeds $\{42, 123\}$ on the OOF predictions, then run Nelder-Mead on the resulting blended logits to find the 18-dimensional threshold vector δ^*. Both w and δ^* are frozen before any blind test inference is performed. The final predicted level for a sentence with logit vector $\hat{\mathbf{y}}$ is:

$$\hat{y} = \sum_{k=0}^{17} \mathbf{1}[\hat{y}_k > \delta_k^*] + 1 \quad (4)$$

6 Results and Discussion

6.1 Internal Validation and Ablation Studies

To evaluate our modeling choices and the contribution of different objective functions during the Development Phase, we conducted an ablation study on the validation set using AraBERTv02 (Antoun et al., 2020). Table 1 summarizes the internal QWK scores across various loss formulations prior to blind test evaluation.

The results highlight a crucial trade-off dictated by the QWK metric. Nominal classification models (Focal Loss (Lin et al., 2018) and Weighted Cross-Entropy) implicitly optimize for exact matches but

Table 1: Performance comparison of different loss objectives on the validation set. Bold rows denote ordinal regression with CORAL loss using ARBERT† and AraBERTv02‡ backbones.

Paradigm	Loss Function	QWK
Nominal Classification	Focal Loss ($\gamma = 2.0$)	0.7958
Nominal Classification	Weighted Cross-Entropy	0.8035
Ordinal Regression	EMD	0.8036
Standard Regression	Mean Absolute Error	0.8228
Ordinal Regression	**CORAL Loss†**	**0.8241**
Ordinal Regression	**CORAL Loss‡**	**0.8242**

†ARBERTv2 ‡AraBERTv02

fail to penalize distant predictions, resulting in lower QWK scores (0.7958 and 0.8035). While EMD (Hou et al., 2017) models cumulative distribution distances, it only achieved 0.8036.

Conversely, treating the problem as a continuous space via Mean Absolute Error provided a highly competitive QWK of 0.8228, as its distance penalty closely mirrors the QWK mechanism. However, MAE rigidly assumes equidistant intervals between levels. Our CORAL framework (Cao et al., 2020) overcomes this and remains superior (0.8242) by learning 18 flexible binary boundaries. To select the primary backbone for our full pipeline, we additionally evaluate ARBERTv2 (Abdul-Mageed et al., 2021) under the identical CORAL objective. ARBERT achieves a validation QWK of 0.8241, marginally below AraBERTv02 (0.8242). Given this near-identical performance, we select AraBERTv02 as the primary backbone for our 5-fold stratified cross-validation pipeline, as it provides a consistent advantage under the same training conditions.

6.2 Official Blind Test Results

In the Testing Phase, we applied the frozen post-processing pipeline to the multi-seed OOF logits of AraBERTv02. All hyperparameters (the multi-seed blend weight and the 18 Nelder-Mead thresholds (Nelder and Mead, 1965)) were finalized exclusively on the OOF logits prior to blind test inference. The blind test set was loaded and predicted exactly once using the frozen model checkpoints and frozen thresholds, yielding an official blind test QWK of **0.837**.

6.3 Error Analysis

An analysis of the out-of-fold confusion matrices revealed a prevalent issue: conservative prediction bias. Because QWK heavily penalizes distant errors, base models learned to safely cluster their predictions around the dataset's majority classes (e.g., Levels 10, 12, and 14) to minimize expected loss, while severely under-predicting the extreme minority classes (e.g., Levels 18 and 19).

The Nelder-Mead threshold optimization partially addressed this bias by allowing each of the 18 binary decision boundaries to shift independently, improving recall for under-predicted levels without requiring any test-set feedback. However, extreme minority classes remain challenging due to their scarcity in the training distribution. Future work could explore integrating explicit class-frequency anchors directly into the ordinal loss function to further improve coverage of edge-case readability levels.

7 Conclusion

In this paper, we described our submitted system for the BAREC Shared Task 2026. By systematically evaluating multiple paradigms, we demonstrated that when the primary evaluation metric is QWK, reframing the Arabic readability assessment as an ordinal regression problem via the CORAL framework significantly outperforms accuracy-driven classification. Our pipeline showed that combining Parameter-Efficient Fine-Tuning with an ordinal hierarchical loss is highly effective. Furthermore, calibrating an 18-dimensional threshold vector via Nelder-Mead exclusively on OOF logits — without any blind test feedback — successfully countered the conservative prediction bias of ordinal models, achieving an official blind test QWK of **0.837**. Future work could explore integrating explicit class-frequency anchors directly into the ordinal loss function to further refine predictions for extreme minority readability levels.

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